

Mathematical Expectation or Expected value of a random variable:-

The average value ^{or expected value} of a random phenomenon is termed as its mathematical expectation.
expression for

The mathematical Expectation of a random variable X with probability mass function (P.m.f) $p(x)$ is given as:-

$$E(X) = \sum x p(x)$$

The Expectation of a random variable X , with probability density function (P.d.f) $f(x)$ is given as

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Expected Value of function of a random variable

Consider a random variable X , with p.m.f $p(x)$ ~~and~~. If $g(\cdot)$ is a function such that $g(X)$ is a r.v ~~and~~ then

$$E(g(X)) = \sum_i g(x_i) p_i = \sum g(x) p(x) \quad (X \text{ is discrete r.v})$$

If X is continuous random variable with p.d.f $f(x)$,

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) f(x) dx$$

Mean, variance and standard deviation of Random Variable (X) :-

Particular case :-

Let $g(x) = x^r$, r being a +ive integer

$$E(x^r) = \int_{-\infty}^{\infty} x^r \cdot f(x) dx \quad (\text{for continuous random variable})$$

$$E(x^r) = \sum x^r \cdot p(x) \quad (\text{for discrete random variable})$$

Mean, variance and standard deviation of Random variable (X) :-

Mean :-

(i) For discrete random variable :-

$$\text{Mean}(X) = E(X) = \sum x p(x)$$

(ii) For continuous random variable :-

$$\text{Mean}(X) = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Variance :- For discrete random variable :-

$$\sigma^2 = \text{Var}(X) = E(X - E(X))^2$$

$$= E(X^2) - [E(X)]^2$$

$$\text{or } \boxed{\sigma^2 = \sum (x - \bar{x})^2 p(x)}$$

For continuous random variable :-

$$\sigma^2 = \int_{-\infty}^{\infty} (x - E(X))^2 \cdot f(x) dx$$

$$\left[\sigma^2 = \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) dx \right]$$

Prop. of Expectation:-

$$E(a) = a$$

(1) $E(ax) = aE(x)$ and $E(ax+b) = aE(x) + b$

(2) $\text{Var}(ax) = a^2 \text{Var}(x)$ and $V(b) = 0$, $b = \text{const.}$

(3) $E(x+y) = E(x) + E(y)$ (Addition theorem of expectation)

(4) ~~If x and y are~~(4) If x and y are independent Random variables then

$$E(xy) = E(x) \cdot E(y) \quad \left\{ \begin{array}{l} \text{Multiplication} \\ \text{theo. of Expectation} \end{array} \right.$$

(5) $E(\frac{1}{x}) \neq \frac{1}{E(x)}$, $E(x^2) \neq [E(x)]^2$, $E(x^{1/2}) \neq (E(x))^{1/2}$

Q:- A r.v. x has the p.d.f $f(x) = \begin{cases} \frac{x+1}{2}, & -1 < x < 1 \\ 0, & \text{o.w.} \end{cases}$

find its mean and variance.

Solⁿ:-Mean of cont. random var. x is given by

$$\text{Mean} = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$\text{Mean} = \int_{-1}^1 x \left(\frac{x+1}{2} \right) dx$$

$$\left[\text{Mean} = \frac{1}{3} \right]$$

$$\text{Variance} = \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) dx$$

$$= \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) dx$$

$$\text{Variance} = \int_{-1}^1 \left(x - \frac{1}{3}\right)^2 \cdot \left(\frac{x+1}{2}\right) dx$$

$$\text{Variance} = \frac{2}{9}$$

Q:- X be a r.v. with p.d.f given by
 $f_X(x) = \frac{1}{2} e^{-|x|}, \forall x \in \mathbb{R}.$

If $Y = X^2$, find c.d.f of Y .

Soln
 as $x \in \mathbb{R}$ but $Y = X^2$
 ie $R_Y = [0, \infty)$ ie $y \in [0, \infty)$

Now c.d.f of Y $F_Y(y) = P(Y \leq y)$

$$F_Y(y) = P(X^2 \leq y)$$

$$= P(-\sqrt{y} \leq X \leq \sqrt{y})$$

$$= \int_{-\sqrt{y}}^{\sqrt{y}} f_X(x) dx$$

$$= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} e^{-|x|} dx$$

$$= \int_0^{\sqrt{y}} e^{-x} dx = 1 - e^{-\sqrt{y}}$$

$$\text{ie } F_Y(y) = \begin{cases} 1 - e^{-\sqrt{y}}, & y \geq 0 \\ 0, & \text{o.w.} \end{cases}$$

Q:- let X be a r.v with p.d.f given by

$$f_X(x) = \begin{cases} c x^2, & |x| \leq 1 \\ 0, & \text{o.w.} \end{cases}$$

- (a) Find c (b) find $E(X)$ and $\text{Var}(X)$
 (c) find $P(X \geq 1/2)$

Soln.: (a) as $\int_{-1}^1 f(x) dx = 1$

$$\int_{-1}^1 c x^2 dx = 1$$

$$\Rightarrow \boxed{c = 3/2}$$

(b) $E(X) = \int_{-1}^1 x \cdot \frac{3}{2} x^2 dx = 0$

$$V(X) = \int_{-1}^1 x^2 f(x) dx = \frac{3}{2} \int_{-1}^1 x^4 dx$$

$$= \frac{3}{2} \cdot \frac{2}{5} = \frac{3}{5}$$

$$V(X) = \frac{3}{5}$$

(c) $P(X \geq 1/2) = \int_{1/2}^1 \frac{3}{2} x^2 dx = \frac{7}{16}$

A random variable X has following probability function:-

x :	0	1	2	3	4	5	6	7
$p(x)$:	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

- (a) find K (b) Find distribution fun of X
 (c) Find mean and Variance

soln

$$K = \frac{1}{10}$$

(b) Distribution function

x :	0	1	2	3	4	5	6	7
$P(X \leq x) = F(x)$:	0	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{5}{10}$	$\frac{8}{10}$	$\frac{81}{100}$	$\frac{83}{100}$	1

(c) Mean(X) = $\sum x p(x)$

$$E(X) = 0 + \frac{1}{10} + \frac{4}{10} + \frac{6}{10} + \frac{12}{10} + \frac{5}{100} + \frac{12}{100} + \frac{119}{100}$$

$$\text{Mean}(X) = 3.66$$

$$\text{Variance} = \sum x^2 p(x) - [\sum x p(x)]^2 = E(X^2) - (E(X))^2$$

$$E(X^2) = \sum x^2 p(x) = \frac{1}{10} + \frac{8}{10} + \frac{18}{10} + \frac{48}{10} + \frac{25}{100} + \frac{72}{100} + \frac{833}{100}$$

$$= 16.8$$

$$E(X) = 3.66$$

$$\text{So Variance} = 16.8 - (3.66)^2$$

$$\text{Variance} = 3.4$$

Q:- A cont. r.v has density function

$$f(x) = \begin{cases} 2e^{-2x}, & x > 0 \\ 0, & \text{o.w.} \end{cases} \quad \text{find expected value \& variance of } X.$$

soln:- $E(X) = \int_0^{\infty} x f(x) dx = \frac{1}{2} \quad \text{Var}(X) = \frac{1}{4}$

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Q Find Mean and Variance of probability density function given as

$x:$	1	2	3	4	5
$p(x):$	0.2	0.35	0.25	.15	.05

Ans $E(x) = 2.5$ Variance $= 1.25$

Two Dimensional Random Variable:-

Let S be sample space. Let X and Y be two functions each assigning a real no. to each outcome $s \in S$, then (X, Y) is a two dimensional random variable, then assigns a real point in R^2 to each outcome.

Discrete R.V

when (X, Y) take a discrete ~~ran~~ values (ie finite) then (X, Y) is called a two dimensional discrete R.V and values of (X, Y) may be represented as (x_i, y_j) when $i=1, 2, \dots, n$; $j=1, 2, \dots, m$

In the study of two dimensional discrete R.V's we have the following 5 important terms:-

- Joint Probability Function (J.P.F) or Joint Probability Mass function
- Joint Probability distribution
- Marginal Probability function of X
- Marginal Probability function of Y
- Conditional Probability function

Continuous Random Variable:-

when (X, Y) can take all the values in a region R , in $X-Y$ plane then (X, Y) is called two-dimensional continuous random variable.

- Joint probability density function
- Joint probability distribution function
- Marginal probability density function
- Conditional probability density function

eg: Consider the experiment of tossing a coin twice. The sample space is

$$S = \{HH, HT, TH, TT\}$$

let X denotes the no. of heads obtained in first toss, Y denotes the no. of heads in second toss then

$S:$	HH	HT	TH	TT
$X(s):$	1	1	0	0
$Y(s)$	1	0	1	0

then range space of (X, Y) is

$\{(1, 1), (1, 0), (0, 1), (0, 0)\}$ which is

finite so (X, Y) is discrete r.v.

Joint Probability mass function:-

If (X, Y) is two dimensional discrete random variable then joint probability mass function of X, Y is defined as

$$P(X=x_i, Y=y_j) = P(x_i, y_j) = p_{ij}$$

It satisfied the following properties:-

(i) $p_{ij} \geq 0, \forall i, j$

(ii) $\sum_j \sum_i p_{ij} = 1$

Note that

$$\begin{aligned} P(X=x_i, Y=y_j) \\ &= P[(X=x_i) \cap (Y=y_j)] \\ &= p_{ij} \end{aligned}$$

Let $X = \{x_1, x_2, \dots, x_n\}$

$Y = \{y_1, y_2, \dots, y_m\}$

$X \times Y = \{x_1, x_2, \dots, x_n\} \times \{y_1, y_2, \dots, y_m\}$

~~XXXX~~ then $P(X=x_i, Y=y_j) = p_{ij}$ can be represented by following table.

$X \backslash Y$	y_1	y_2	y_3	...	y_m	Total
x_1	p_{11}	p_{12}	p_{13}		p_{1m}	$p_{1\cdot}$
x_2	p_{21}	p_{22}	p_{23}		p_{2m}	$p_{2\cdot}$
x_3						
$\vdots$						
$\vdots$						
$\vdots$						
x_n	p_{n1}	p_{n2}	p_{n3}		p_{nm}	$p_{n\cdot}$
Total	$p_{\cdot 1}$	$p_{\cdot 2}$	$p_{\cdot 3}$		$p_{\cdot m}$	1

Figure-1

Q:- Two fair dice are thrown. Let X_1 be the score on first die and X_2 be the score on the second die. Let Y denote the maximum of X_1 and X_2 i.e. $Y = \max(X_1, X_2)$. Find the joint probability distribution of Y and X_1 .

Solⁿ Each random variable X_1 and X_2 take six values 1, 2, 3, 4, 5, 6

$$P(X_1 = i) = \frac{1}{6}, \quad P(X_2 = i) = \frac{1}{6}, \quad i = 1, 2, 3, 4, 5, 6$$

$$Y = \max(X_1, X_2)$$

Obv. 1) $P(X_1 = i, Y = j) = 0$, when $j < i$

Now 2) $P(X_1 = i, Y = j) = P(X_1 = i) \cdot P(Y = j)$,
where $j > i$

$$= \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

$$P(X_1 = i, Y = j) = \frac{1}{36}, \text{ when } j > i$$

(3) Now

$$P(X_1 = i, Y = i) = P(X_1 = i, X_2 \leq i)$$

$$= P((X_1 = i) \cap (X_2 \leq i))$$

$$= \sum_{j=1}^i P((X_1 = i) \cap (X_2 = j))$$

$$= \sum_{j=1}^i P(X_1 = i) \cdot P(X_2 = j)$$

$$= \sum_{j=1}^i \frac{1}{6} \cdot \frac{1}{6} = \frac{i}{36}, \quad i = 1, 2, \dots, 6$$

Joint Prob. distribution

$x_i \backslash y_j$	1	2	3	4	5	6	T
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$6/36$
2	0	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$6/36$
3	0	0	$\frac{3}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$6/36$
4	0	0	0	$\frac{4}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$6/36$
5	0	0	0	0	$\frac{5}{36}$	$\frac{1}{36}$	$6/36$
6	0	0	0	0	0	$\frac{6}{36}$	$6/36$
T	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$	1

here ① $p_{ij} \geq 0$

$$\textcircled{2} \sum_{j=1}^6 \sum_{i=1}^6 p_{ij} = 1$$

Joint Probability density function:-

If (x, y) is two dimensional continuous random variable then $f(x, y)$ is called joint probability density function, satisfy the following condn:-

① $f(x, y) \geq 0, \quad \forall (x, y) \in R, \quad R \text{ is the range space}$

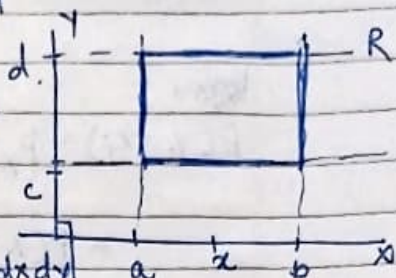
② $\int \int_R f(x, y) dx dy = 1$

ie if D is a subspace of range space R
such that $D: a \leq x \leq b, c \leq y \leq d$

then

$$P((x, y) \in D) = \iint_D f(x, y) dx dy$$

$$\text{ie } \left[P(a \leq x \leq b, c \leq y \leq d) = \int_c^d \int_a^b f(x, y) dx dy \right]$$



Commulation distribution function:-

If (X, Y) is two dimensional R.V then

$F(x, y) = P(X \leq x \text{ and } Y \leq y)$ is called
c.d.f of (X, Y) .

For discrete random variable:-

$$F(x, y) = \sum_{y_i \leq y} \sum_{x_j \leq x} p_{ij}$$

For continuous random variable:-

$$F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(x, y) dx dy$$

Prop. of $F(x, y)$

$$(1) F(-\infty, y) = F(x, -\infty) = 0$$

$$F(\infty, \infty) = 1$$

$$(2) P(a < X < b, Y \leq y) = \int_{-\infty}^y \int_a^b f(x, y) dx dy$$

$$(3) 0 \leq F(x, y) \leq 1$$

(4) $F(x, y)$ is non decreasing function of (x, y)

Marginal Probability function:- (discrete)

~~begin~~

$$P(X=x_i) = p_x(x_i)$$

$$= P(X=x_i \cap Y=y_1) + P(X=x_i \cap Y=y_2) + \dots + P(X=x_i \cap Y=y_m)$$

$$p_x(x_i) = p_{i1} + p_{i2} + p_{i3} + \dots + p_{im} = p_{i.}$$

$$p_x(x_i) = \sum_{j=1}^m p_{ij} \quad (\text{See figure 1})$$

$\leftarrow$ Marginal prob. fun of x .

Now

$$\left[\begin{aligned} \sum_{i=1}^n p_{i.} &= p_{1.} + p_{2.} + p_{3.} + \dots + p_{n.} \\ \sum_{i=1}^n p_{i.} &= \sum_{i=1}^n \sum_{j=1}^m p_{ij} = 1 \end{aligned} \right]$$

Similarly $P(Y=y_j) = p_y(y_j) = p_{.j}$ is called

$$p_y(y_j) = \sum_{i=1}^n p_{ij} \quad \text{Marginal prob of } y.$$

marginal distribution function of y .
Probability

Marginal Probability density function:- (continuous case)

The marginal P.d.f of x, y are denoted by

$f_x(x)$ and $f_y(y)$ represented as

$$f_x(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$P(a \leq X \leq b) = P(a \leq X \leq b, -\infty \leq Y \leq \infty) = \int_a^b \left(\int_{-\infty}^{\infty} f(x, y) dy \right) dx = \int_a^b f_x(x) dx$$

$$P(c \leq Y \leq d) = \int_c^d f_y(y) dy$$

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In case of two dimensional continuous Random Variable Marginal commulative distribution function is given by

$$P(X \leq x) = F_x(x) = \int_{-\infty}^x f_x(x) dx$$

$$P(Y \leq y) = F_y(y) = \int_{-\infty}^y f_y(y) dy$$

Ex:- Ex for Marginal probability function :-

In case of pair of dice.

$$P(X=1) = P(X=1, Y=1) + P(X=1, Y=2) + P(X=1, Y=3) + P(X=1, Y=4) + P(X=1, Y=5) + P(X=1, Y=6)$$

$$= \frac{1}{6} + 0$$

$$P(X=1) = \frac{1}{6}$$

marginal prob. function at $x=1$, $P(X=1) = \frac{1}{6}$

For Discrete Random Variable:-

Marginal commulative distribution function

$$P(X \leq x) = \sum_y P(X=x, Y=y)$$

$$P(Y \leq y) = \sum_x P(X=x, Y \leq y)$$

$P(X/Y)$ means Prob. of occurring X when Y has already been occurred. classmate
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Conditional Probability function:-

Let (x, y) be a discrete two dimensional random variable, then the conditional discrete probability function or conditional probability mass function of X , given $Y=y$.

$$P_{X/Y}(x/y) = \frac{P(X=x, Y=y)}{P(Y=y)} \quad \text{provided } P(Y=y) \neq 0$$

conditional prob:-

$$\text{or } P(X=x_i / Y=y_j) = \frac{P(X=x_i, Y=y_j)}{P(Y=y_j)} = \frac{p_{ij}}{p_j}$$

Similarly

$$P_{Y/X}(y/x) = \frac{P(X=x, Y=y)}{P(X=x)} \quad , \quad P(X=x) \neq 0$$

Cond. prob is

$$\text{i.e. } P(X=x_i, Y=y_j) = \frac{P(X=x_i, Y=y_j)}{P(X=x_i)} = \frac{p_{ij}}{p_i}$$

The collection of pairs $\{x_i, \frac{p_{ij}}{p_j}\}$ $i=1, 2, \dots$ is called conditional prob. ~~distribution~~ of X given $Y=y_j$ $= y$ function.

Similarly

The collection of pairs $\{y_j, \frac{p_{ij}}{p_i}\}$ $j=1, 2, \dots$ is called condition. probability ~~dist~~ of Y given $X=x_i$ function.

Similarly for continuous case:-

1. $f_{x/y}(x/y) = \frac{f(x, y)}{f_y(y)}$ is called conditional

probability density function of x , given y .
denoted by $f(x/y)$

2. $f_{y/x}(y/x) = \frac{f(x, y)}{f_x(x)}$, is called conditional

probability density function of y , given x .
denoted by $f(y/x)$.

Independent random variables:-

If (x, y) is two dimensional independent R.Vs. then

$$P(X=x, Y=y) = P(X=x) \cdot P(Y=y) \quad \text{Discrete}$$

$$f(x, y) = f(x) \cdot f(y) \quad \text{continuous.}$$

Q-1 Joint Probability distribution of two R.V. X and Y is given below

X \ Y	1	2	3	4	T
1	$4/36$	$3/36$	$2/36$	$1/36$	$10/36$
2	$1/36$	$3/36$	$3/36$	$2/36$	$9/36$
3	$5/36$	$1/36$	$1/36$	$1/36$	$8/36$
4	$1/36$	$2/36$	$1/36$	$5/36$	$9/36$
T	$11/36$	$9/36$	$7/36$	$9/36$	1

① Find Marginal distribution of X and Y.

② conditional distribution of X given the Y=1 and that of Y given the value of X=2.

Soln ① -

X	1	2	3	4
P(X)	$10/36$	$9/36$	$8/36$	$9/36$

Y	1	2	3	4
P(Y)	$11/36$	$9/36$	$7/36$	$9/36$

$$\textcircled{2} \quad P(X=x | Y=1) = \frac{P(X=x, Y=1)}{P(Y=1)}$$

$$P(X=1 | Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{4/36}{11/36} = \frac{4}{11}$$

$$P(X=2 | Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{1/36}{11/36} = \frac{1}{11}$$

$$P(X=3 | Y=1) = \frac{P(X=3, Y=1)}{P(Y=1)} = \frac{2/36}{11/36} = \frac{2}{11}$$

$$P(X=4 | Y=1) = \frac{1}{11}$$

conditional prob. distribution of x , given $y=1$.

x	1	2	3	4
$P(x=x y=1)$	$4/11$	$1/11$	$5/11$	$1/11$

conditional probability function of y given $x=2$:-

$$P(Y=y|X=2) = \frac{P(X=2, Y=y)}{P(X=2)}$$

$$P(Y=1|X=2) = \frac{P(X=2, Y=1)}{P(X=2)} = \frac{1/36}{9/36} = \frac{1}{9}$$

$$P(Y=2|X=2) = \frac{P(X=2, Y=2)}{P(X=2)} = \frac{3/36}{9/36} = \frac{1}{3}$$

$$P(Y=3|X=2) = \frac{P(X=2, Y=3)}{P(X=2)} = \frac{3/36}{9/36} = \frac{1}{3}$$

$$P(Y=4|X=2) = \frac{P(X=2, Y=4)}{P(X=2)} = \frac{2/36}{9/36} = \frac{2}{9}$$

conditional distribution of y given $x=2$

$y=y$	1	2	3	4
$P(Y=y X=2)$	$1/9$	$1/3$	$1/3$	$2/9$

Ex:- For the bivariate probability distribution
 X & Y find

- (1) $P(X \leq 1, Y = 2)$ (2) $P(X \leq 1)$ (3) $P(Y \leq 3)$
 (4) $P(X < 3, Y \leq 4)$

$X \backslash Y$	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{2}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

Solution:- complete the table by writing the marginal prob. distribution

$X \backslash Y$	1	2	3	4	5	6	$P_X(X)$
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{2}{32}$	$\frac{8}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{10}{16}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$	$\frac{8}{64}$
$P_Y(Y)$	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$	1

$$(1) P(X \leq 1, Y = 2) = P(X = 0, Y = 2) + P(X = 1, Y = 2) \\ = 0 + \frac{1}{16} = \frac{1}{16}$$

$$(2) P(X \leq 1) = P(X = 0) + P(X = 1) \\ = \frac{8}{32} + \frac{10}{16} = \frac{28}{32}$$

$$(3) P(Y \leq 3) = P(Y = 1) + P(Y = 2) + P(Y = 3) \\ = \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{23}{64}$$

$$\textcircled{4} \quad P(X < 3, Y \leq 4) = P(X=0, Y \leq 4) + P(X=1, Y \leq 4) + P(X=2, Y \leq 4)$$

$$= \left(\frac{1}{32} + \frac{2}{32} \right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{8} \right) + \left(\frac{1}{32} + \frac{1}{32} + \frac{1}{64} + \frac{1}{64} \right)$$

$$= \frac{3}{32} + \frac{6}{16} + \frac{6}{64} = \frac{36}{64}$$

Q:- The joint PMF of X & Y is given below. Find the marginal p.m.f for X & Y .

Soln:-

$P_{XY}(x,y)$	$y=0$	$y=1$	$y=2$
$x=0$	0.01	0	0
$x=1$	0.09	0.09	0
$x=2$	0	0	0.81

Find the marginal p.m.f for the x -vs Y and X .

Soln:-

$x:$	0	1	2
$P_X(x):$	0.1	0.18	0.81

$y:$	0	1	2
$P_Y(y):$	0.10	0.09	0.81

Q For a bivariate prob. distribution given below.

$x \backslash y$	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

find (a) $P(Y \leq 3 | X \leq 1)$ (b) $P(X + Y \leq 4)$

Soln.: (a) $P(Y \leq 3 | X \leq 1) = \frac{P(X \leq 1, Y \leq 3)}{P(X \leq 1)}$

$$= \frac{P(X=0, Y \leq 3) + P(X=1, Y \leq 3)}{P(X \leq 1)}$$

$$= \frac{\frac{1}{32} + \frac{4}{16}}{\frac{8}{32} + \frac{20}{32}} = \frac{\frac{9}{32}}{\frac{28}{32}} = \frac{9}{28}$$

$$\begin{aligned} \text{(b) } P(X+Y \leq 4) &= \sum_{j=1}^4 P(X=0, Y=j) + \sum_{j=1}^3 P(X=1, Y=j) \\ &\quad + \sum_{j=1}^2 P(X=2, Y=j) \\ &= \frac{3}{32} + \frac{1}{4} + \frac{1}{16} = \frac{13}{32} \end{aligned}$$

8:- Three balls are drawn at random without replacement from a box containing 2 white, 3 red, and 4 black balls. If x denotes the no. of white ball drawn and y denote the no. of red balls drawn, find the joint probability distribution of (x, y) .

Solⁿ Here x takes the values
0, 1, 2

y takes the values
0, 1, 2, 3

2 - white

3 - Red

4 - Black.

9 balls

Now. As 3 balls are to be drawn.

x - no. of white

$P(x=0, y=0) = \text{Prob. of getting 0 red and 0 white ball.}$

y - no. of red.

$$= P(\text{All 3 black balls})$$

$$= \frac{{}^4C_3}{{}^9C_3} = \frac{1}{21}$$

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$$P(X=0, Y=1) = P(\text{getting of } \cancel{2} \text{ 1 red and 2 black})$$

$$= \frac{{}^3C_1 \times {}^4C_2}{{}^9C_3}$$

$$P(X=0, Y=1) = \frac{3}{14}$$

slly $P(X=0, Y=2) = \frac{{}^3C_2 \times {}^4C_1}{{}^9C_3}$

$$P(X=0, Y=2) = \frac{1}{7}$$

slly other probabilities can be found.

Joint probability distribution of (X, Y) is given by

$X \backslash Y$	0	1	2	3
0	$\frac{1}{21}$	$\frac{3}{14}$	$\frac{1}{7}$	$\frac{1}{84}$
1	$\frac{1}{7}$	$\frac{2}{7}$	$\frac{1}{14}$	0
2	$\frac{1}{21}$	$\frac{1}{28}$	0	0